

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: ITA 06

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO2: *Neural Network Representation, Perceptron Model, Multilayer Perceptrons, Backpropagation Algorithm, Deep Neural Networks, Genetic Algorithms, Hypothesis Space Search, Genetic Programming, Models of Evaluation and Learning (BL1, BL2)*

Assessment Tool Weightage: 20%

QUESTIONS: 15

Total Marks:25

Annexure A

SCENARIO-BASED QUESTIONS

Code of Conduct:

I GAYATHRI N.P (192421362) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Gayathri N P

Signature of the student:

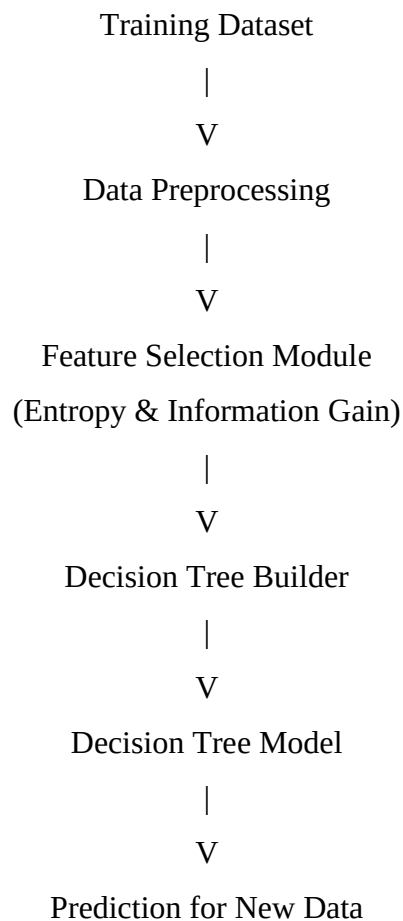
Annexure A

6. With the help of a neat diagram, explain the architecture of a decision tree learning system. Describe how features are selected using measures such as entropy and information gain. Illustrate the process of tree construction and pruning using a small example dataset, and explain how overfitting can be controlled in decision tree models.

Definition

A Decision Tree is a supervised machine learning algorithm used for classification and regression. It predicts the output by splitting the dataset into branches based on feature values.

Architecture Diagram



Feature Selection

The best feature is selected using **Entropy** and **Information Gain**.

Entropy Formula:

$$\text{Entropy}(S) = -\sum p_i \times \log_2(p_i)$$

Where:

- p_i = Probability of each class

Information Gain Formula:

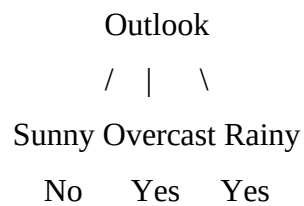
$$\text{Information Gain} = \text{Entropy}(\text{Parent}) - \sum (|\text{Child}| / |\text{Parent}|) \times \text{Entropy}(\text{Child})$$

The attribute with the **highest information gain** is chosen as the splitting attribute.

Example Dataset

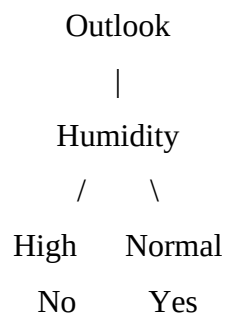
Outlook	Play
Sunny	No
Rainy	Yes
Sunny	No
Overcast	Yes

Decision Tree

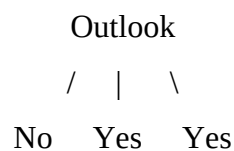


Pruning

Before Pruning



After Pruning



Controlling Overfitting

- Pre-pruning
- Post-pruning
- Limit tree depth

- Minimum samples per node
- Cross-validation

Advantages

- Easy to understand
- Fast prediction
- Handles both numerical and categorical data

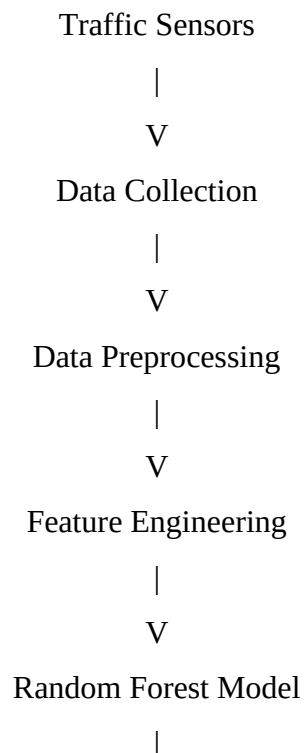
7. Design a machine learning system for smart traffic management to predict traffic congestion using data such as vehicle density, time of day, weather conditions, and road type. Justify the choice of algorithm, explain preprocessing and feature engineering steps, and describe how the model can be trained, validated, and deployed for real-time decision-making.

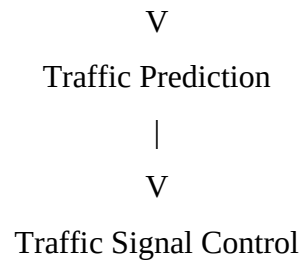
Objective

Predict traffic congestion using:

- Vehicle Density
- Time of Day
- Weather Conditions
- Road Type

System Architecture





Choice of Algorithm

Random Forest is selected because it:

- Gives high accuracy
- Handles missing values
- Reduces overfitting
- Works well with multiple features

Data Preprocessing

- Remove missing values
- Remove duplicate records
- Normalize numerical values
- Encode categorical variables
- Split data into training and testing sets

Feature Engineering

Important features include:

- Vehicle Density
- Time of Day
- Weather
- Road Type
- Day of Week

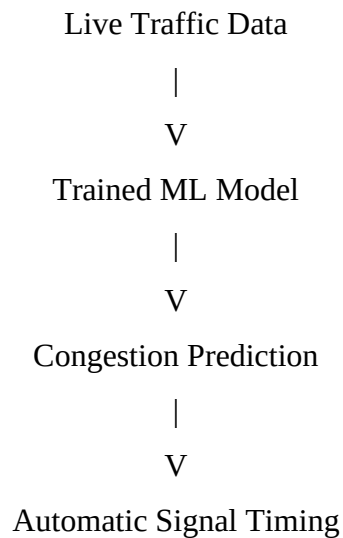
Model Training

- Train the Random Forest model
- Tune hyperparameters
- Validate using testing data

Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1-score

Deployment



Advantages

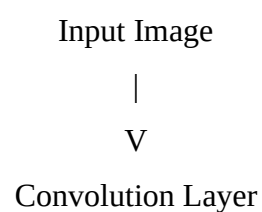
- Reduces traffic congestion
- Saves fuel
- Improves emergency response
- Supports real-time decisions

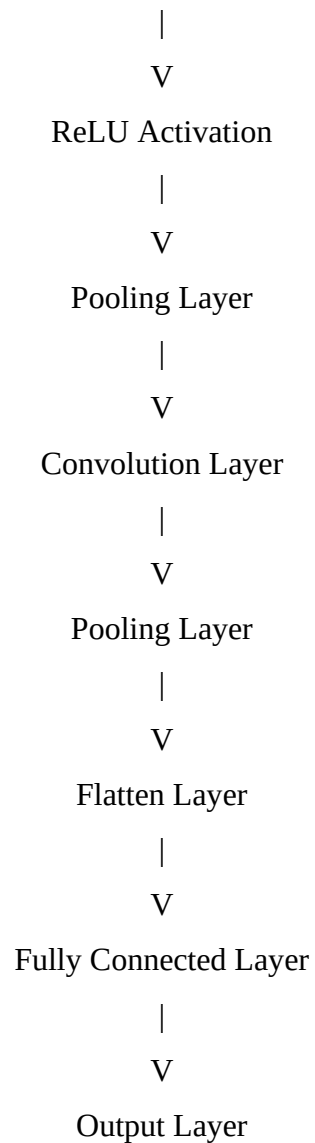
8. In the context of Convolutional Neural Networks (CNNs), explain the different layers involved, including convolutional, pooling, and fully connected layers. Describe how feature maps are generated and how backpropagation updates weights in CNNs. Illustrate the process with an example of image classification and discuss factors affecting model performance.

Definition

A Convolutional Neural Network (CNN) is a deep learning model mainly used for image classification and object detection.

CNN Architecture





Layers

1. Convolution Layer

- Applies filters to detect features such as edges and textures.
- Produces feature maps.

Feature Map = Input Image * Filter

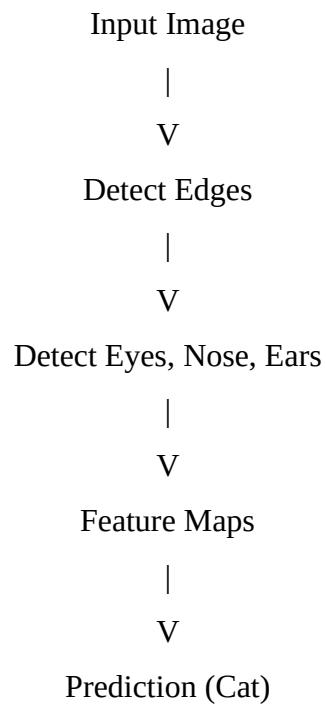
2. Pooling Layer

- Reduces image size.
- Types:
 - Max Pooling
 - Average Pooling
- Reduces computation and overfitting.

3. Fully Connected Layer

- Combines extracted features.
- Produces the final classification.

Feature Map Example



Backpropagation

1. Calculate prediction error.
2. Compute gradients.
3. Update weights.

Weight Update Formula:

$\text{New Weight} = \text{Old Weight} - \text{Learning Rate} \times \text{Gradient}$

Factors Affecting Performance

- Dataset size
- Learning rate
- Number of filters
- Epochs
- Batch size
- Data augmentation

Applications

- Face Recognition
- Medical Imaging
- Self-driving Cars
- Object Detection

9. Consider the application of Genetic Algorithms for feature selection in a machine learning problem. Explain how candidate feature subsets are represented as chromosomes and how fitness is evaluated. Describe the role of selection, crossover, and mutation in evolving better feature subsets, and discuss how this improves model accuracy and reduces complexity.

A Genetic Algorithm (GA) is an optimization technique inspired by natural evolution. It selects the best subset of features for machine learning.

Chromosome Representation

Features:

F1 F2 F3 F4 F5

Chromosome:

1 0 1 1 0

Selected Features:

- F1
- F3
- F4

Genetic Algorithm Process

Initial Population

|

V

Fitness Evaluation

|

V

Selection

|

V

Crossover

|

V

Mutation

|

V

New Population

|
V
Best Feature Subset

Fitness Evaluation

Fitness is measured using:

- Classification Accuracy
- Number of Selected Features

Higher fitness indicates a better solution.

Selection

Best chromosomes are selected for reproduction.

Methods:

- Roulette Wheel Selection
- Tournament Selection

Crossover Example

Parent 1

11010

Parent 2

00111

Child

11111

Mutation Example

Before Mutation

11010

After Mutation

11000

Advantages

- Removes unnecessary features
- Improves model accuracy
- Reduces computational cost
- Prevents overfitting

10. In the context of classification problems, explain how different evaluation metrics such as accuracy, precision, recall, F1-score, and ROC curve are used to assess model performance. Using a sample confusion matrix, demonstrate the calculation of these metrics and discuss their importance in applications such as medical diagnosis and fraud detection.

Definition

Evaluation metrics measure the performance of classification models.

Confusion Matrix

Actual / Predicted	Positive	Negative
Positive	TP = 40	FN = 10
Negative	FP = 5	TN = 45

1. Accuracy

Formula:

$$\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN)$$

Calculation:

$$\text{Accuracy} = (40 + 45) / (40 + 45 + 5 + 10)$$

$$\text{Accuracy} = 85 / 100 = \mathbf{85\%}$$

2. Precision

Formula:

$$\text{Precision} = TP / (TP + FP)$$

Calculation:

$$\text{Precision} = 40 / (40 + 5)$$

$$\text{Precision} = 40 / 45 = \mathbf{88.9\%}$$

3. Recall

Formula:

$$\text{Recall} = TP / (TP + FN)$$

Calculation:

$$\text{Recall} = 40 / (40 + 10)$$

$$\text{Recall} = 40 / 50 = \mathbf{80\%}$$

4. F1-Score

Formula:

$$\text{F1-Score} = (2 \times \text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

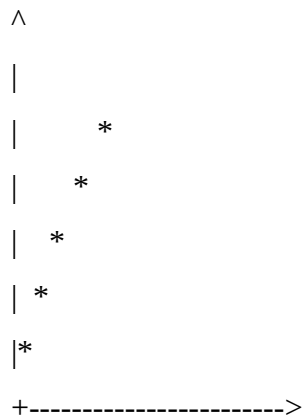
Calculation:

$$\text{F1-Score} = (2 \times 0.889 \times 0.80) / (0.889 + 0.80)$$

$$\text{F1-Score} = \mathbf{84.2\%}$$

5. ROC Curve

True Positive Rate



False Positive Rate

A larger Area Under the Curve (AUC) indicates better model performance.

Importance

Medical Diagnosis

- Recall is more important because missing a disease case can be dangerous.

Fraud Detection

- Precision is more important because it reduces false fraud alerts.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	20	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	15	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand